

PUSHKRAJ KOHOK

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EDUCATION

Illinois Institute of Technology, Chicago, IL || **MS in Artificial Intelligence** August 2024 – April 2026
Coursework: Advanced AI, ML, Deep Learning, NLP, Computer Vision, Advanced Algorithms, Psychology

VIIT, Pune, India || **Bachelor of Computer Engineering (Data Science Major)** June 2016 – May 2020
Coursework: AI and Robotics, Intro to ML, Data Analytics, Adv. Data Structures, Adv. Mathematics
Publication: International Journal of Advance Science & Technology by **Scopus** ([Pg No-1191](#))

WORK EXPERIENCE

Full-Stack AI Engineer (Volunteer) August 2025 – October 2025
Stealth Startup (The Vault) Chicago, USA

- Eliminated DA effort by **90%** by building dashboard with RAG-powered dynamic reports, & smart sales trends.
- Reduced transaction time by **80%** & increased profits, by developing an **instant-sell feature**.
- Delivered **10x** faster search results, by accelerating search through AI-backed vector-DB integration.

Machine Learning Engineer January 2023 – June 2023
Stealth Startup Pune, India

- **Increased precision by 20%** by reducing misclassification of the data detection, by improvising a hybrid ML models for page classification using Python and C#.
- Addressed over 50 business critical bugs, **increasing the accuracy by around 70%** in Data extraction.
- Trained ML models using Form Recognizer, to fix incorrect data extraction for unique formats of pages.

Full-Stack Software Engineer September 2022 – December 2022
Freelance Pune, India

- Achieved a **300% increase in sales** within 1 month by developing a Python-based automation model for a startup's digital marketing processes, leveraging Gmail SMTP and custom HTML design.
- **Led** a team of **3 interns**, to develop an e-commerce website for a shipping business using WordPress.

Full-Stack Software Engineer October 2020 – April 2022
Druva Data Solutions Pvt. Ltd. Pune, India

- **Improved efficiency by 16%** by optimizing data pipeline and APIs using Python (Flask); and resolved legacy bugs.
- **Ensured 100% reliability** of the main product branch by Owning StorageConfig automation and managing Jenkins pipelines for daily cloud pushes which **ran 150+ scheduled tests**, aligned with left-shift SDLC principles.
- **Reduced 2 weeks of Manual Testing efforts by Automating** test cases in **Python** using **pytest** framework for StorageConfig and stats of Global Dashboard. Created Test cases and Test Plans for multiple features.
- **Docker** and **AWS** cloud deployment and debugging using EC2 instances. Practiced Agile Methodology.
- Executed DEV testing to increase code quality as part of team agreement by cross-team collaboration.

Research Assistant – Emotion Analysis Project January 2019 – May 2019
Vishwakarma Institute of Information and Technology Pune, India

- **Improved** emotion recognition **accuracy by 10%** on professor's published research by refining MFCCs via Praat.
- Optimal emotion classification achieved with a tested **SVM** and **Random Forest** combination.

PROJECTS

TeachFusion-AI | RAG, multi-agent teaching platform, vector search [link](#)

- Improved lesson-planning efficiency by **30%** by generating grounded course plans from uploaded materials and reducing blind AI use in teaching.
- Built with **Gemini APIs, course-material chunking, & 3-agent orchestration** for retrieval-based plan generation.

MedOrbit | Healthcare AI platform, multi-agent clinical orchestration [link](#)

- Ensured **100% doctor-approved** patient communication by transforming consultation transcripts into structured clinical and behavioral insights before patient delivery.
- Built with **React, TypeScript, FastAPI, SQLAlchemy, chunked transcript storage, JWT/RBAC, and multi agent orchestration** for auditable clinical intelligence generation.

CodeBleed | Graph-based repository threat intelligence, AI security analysis [link](#)

- Surfaced **4 key risk signals**—secrets, dependencies, endpoints, and attack paths—by converting repository data into explainable graph-based threat intelligence for faster security triage.
- Built with **FastAPI, Neo4j, GitHub ingestion, and vis.js** for graph-driven attack-path analysis.

SKILLS

Framework: Flask, pytest, Gorilla, Pandas, Scikit-learn, Matplotlib, Numpy, Seaborn, Pytorch, Tensorflow, Keras
Prog. Lang.: Python, Golang, C++, SQL, Typescript, NextJS
Cloud & DevOps: AWS (EC2, S3), Docker, Jenkins, GitLab
Core Competencies: Gen AI, Distributed Systems, Data Structures

Prompt Eng: ChatGPT, LLMs, LangChain, etc.
Tools: MS Office, Virtual Machines
Spoken Languages: English, Spanish, Hindi